

# Calculus II

## Lecture 11

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**Faculty of Engineering and Architecture**

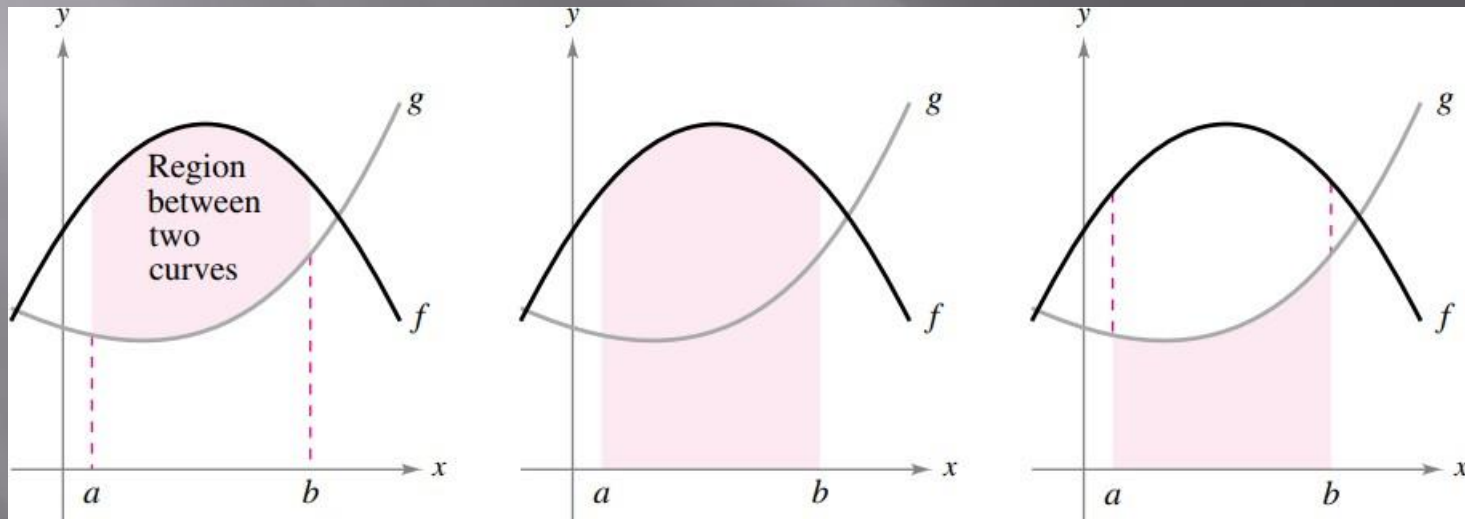
**Fall 2024**

# Areas Between Curves (2/8)

## Area Between two Curves $f(x) \geq g(x)$ :

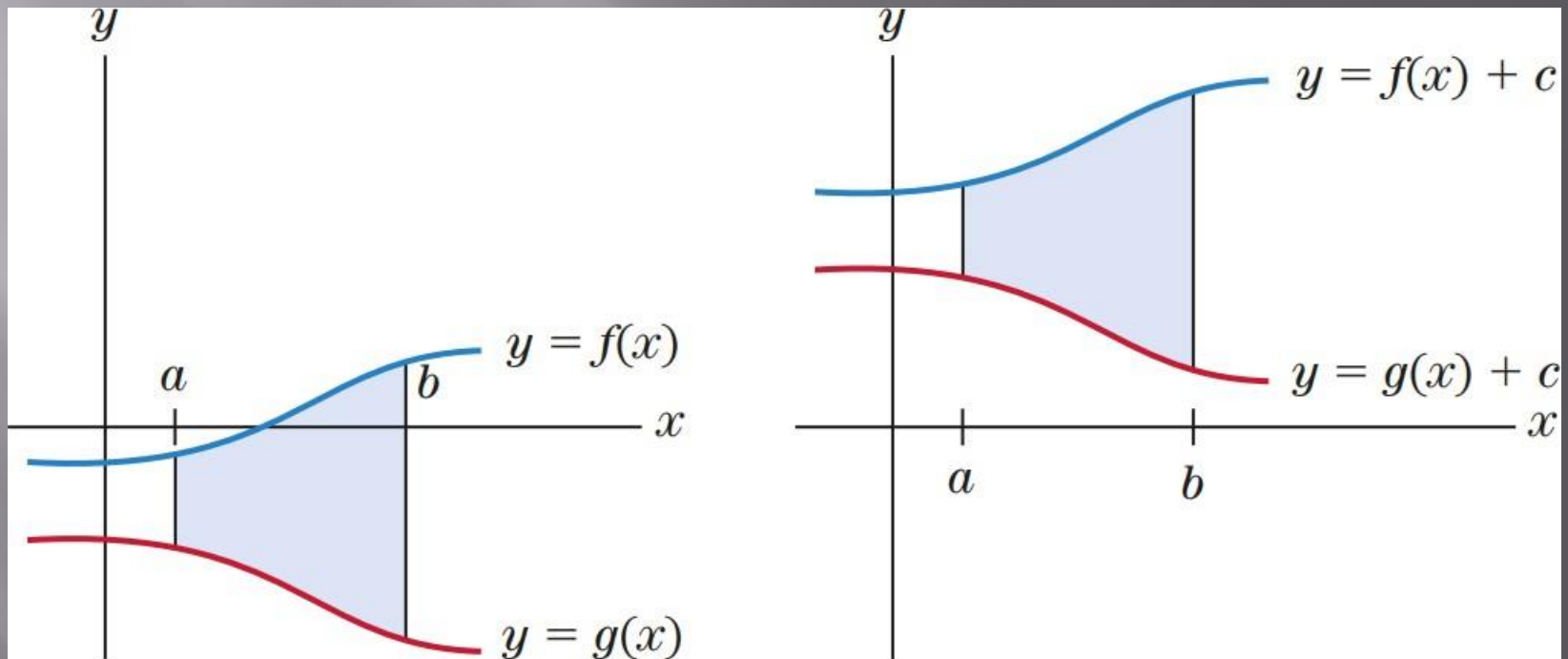
If  $f$  and  $g$  are continuous functions and  $f(x) \geq g(x)$  on  $[a, b]$ , then the area between the curves  $f(x)$  and  $g(x)$  from  $x = a$  to  $x = b$  is given by

$$\int_a^b [f(x) - g(x)] dx.$$



# AREAS BETWEEN CURVES (3/8)

**Shifting a region vertically does not change its area:**

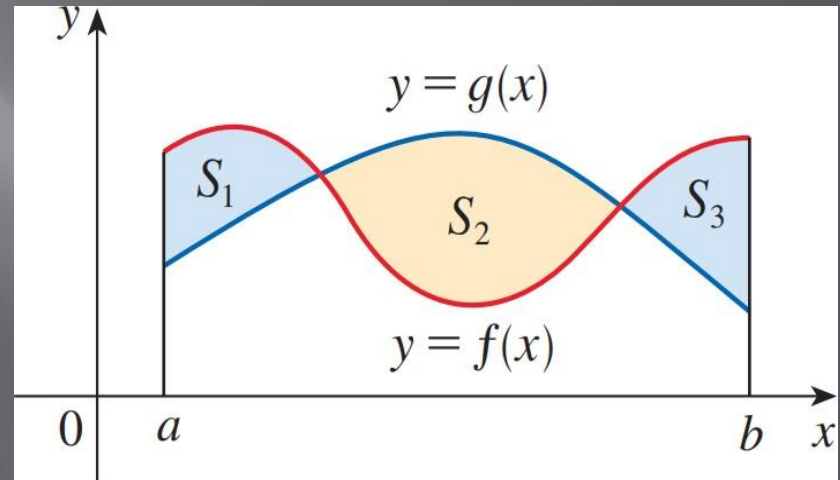


# Areas Between Curves (4/8)

## Area Between two Curves:

If  $f$  and  $g$  are continuous functions on  $[a, b]$ , then the area between the curves  $f(x)$  and  $g(x)$  from  $x = a$  to  $x = b$  is given by

$$A = \int_a^b |f(x) - g(x)| dx$$

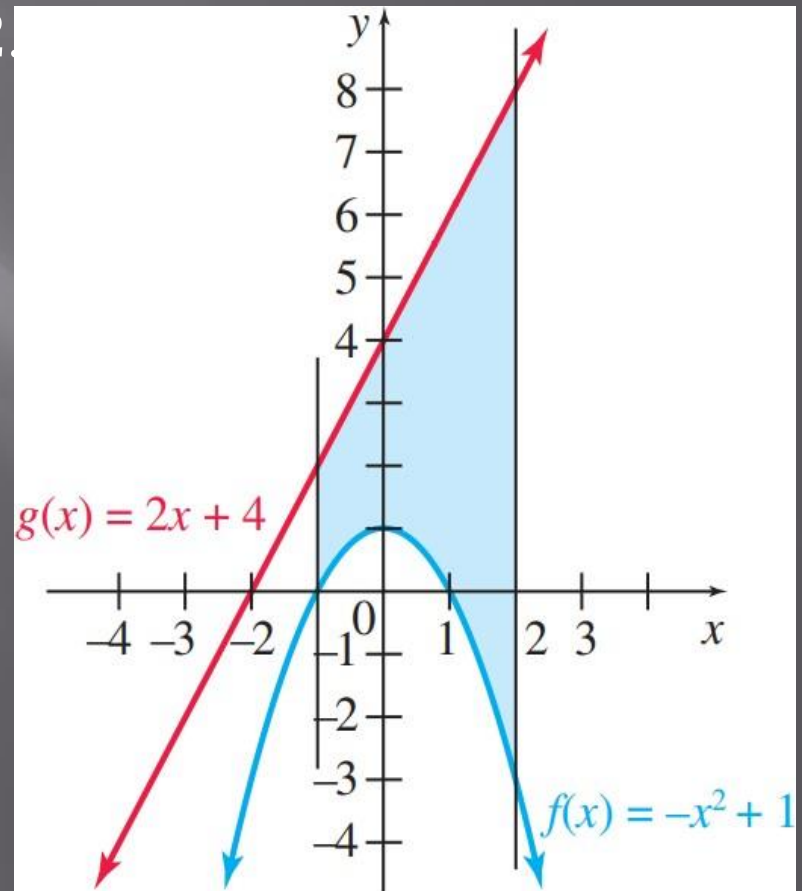


$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{when } f(x) \geq g(x) \\ g(x) - f(x) & \text{when } g(x) \geq f(x) \end{cases}$$

# Areas Between Curves (5/8)

## Example 1 (1/3)

Find the area bounded by  $f(x) = -x^2 + 1$ ,  
 $g(x) = 2x + 4$  from  $x = -1$  to  $x = 2$ .

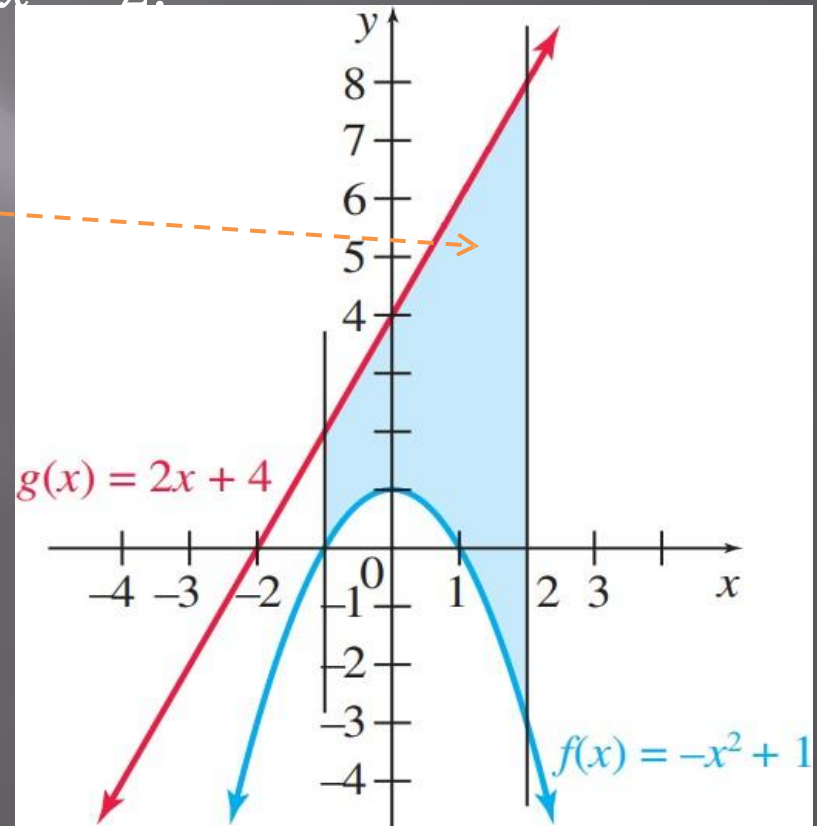


# Areas Between Curves (5/8)

## Example 1 (1/3)

Find the area bounded by  $f(x) = -x^2 + 1$ ,  
 $g(x) = 2x + 4$  from  $x = -1$  to  $x = 2$ .

$$\int_{-1}^2 [g(x) - f(x)] dx$$



# Areas Between Curves (5/8)

## Example 1 (2/3)

Find the area bounded by  $f(x) = -x^2 + 1$ ,  $g(x) = 2x + 4$  from  $x = -1$  to  $x = 2$ .

$$\begin{aligned}\int_{-1}^2 [g(x) - f(x)] dx &= \int_{-1}^2 [(2x + 4) - (-x^2 + 1)] dx \\ &= \int_{-1}^2 (2x + 4 + x^2 - 1) dx \\ &= \int_{-1}^2 (x^2 + 2x + 3) dx\end{aligned}$$

# Areas Between Curves (5/8)

## Example 1 (3/3)

Find the area bounded by  $f(x) = -x^2 + 1$ ,  $g(x) = 2x + 4$  from  $x = -1$  to  $x = 2$ .

$$\int_{-1}^2 (x^2 + 2x + 3) dx$$

$$\begin{aligned} &= \left. \frac{x^3}{3} + x^2 + 3x \right|_{-1}^2 \\ &= \left( \frac{8}{3} + 4 + 6 \right) - \left( \frac{-1}{3} + 1 - 3 \right) \\ &= \frac{8}{3} + 10 + \frac{1}{3} + 2 \\ &= 15. \end{aligned}$$



# Areas Between Curves (6/8)

## Example 2 (1/4)

Find the area between the curves  $f(x) = \sqrt{x}$  and  $g(x) = x^3$

# Areas Between Curves (6/8)

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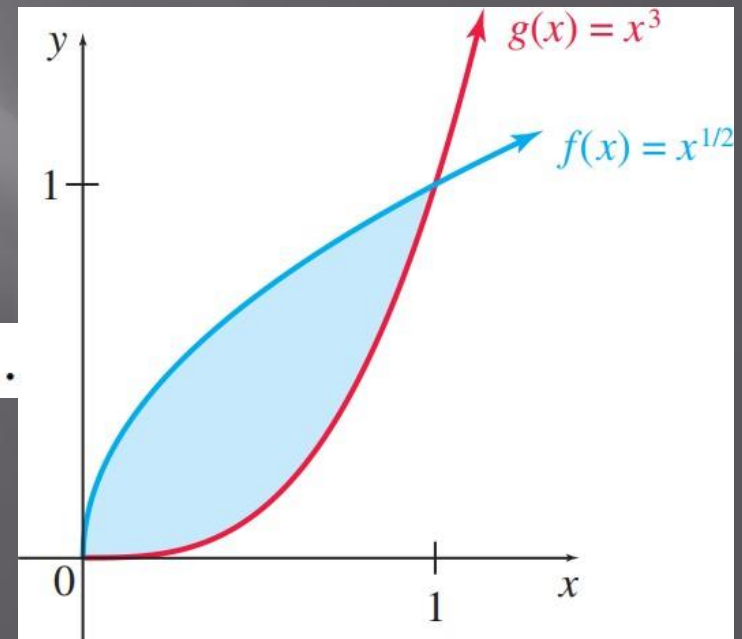
Set the two equal to find where they intersect.

$$x^{1/2} = x^3$$

$$0 = x^3 - x^{1/2}$$

$$0 = x^{1/2}(x^{5/2} - 1)$$

The only solutions are  $x = 0$  and  $x = 1$ .

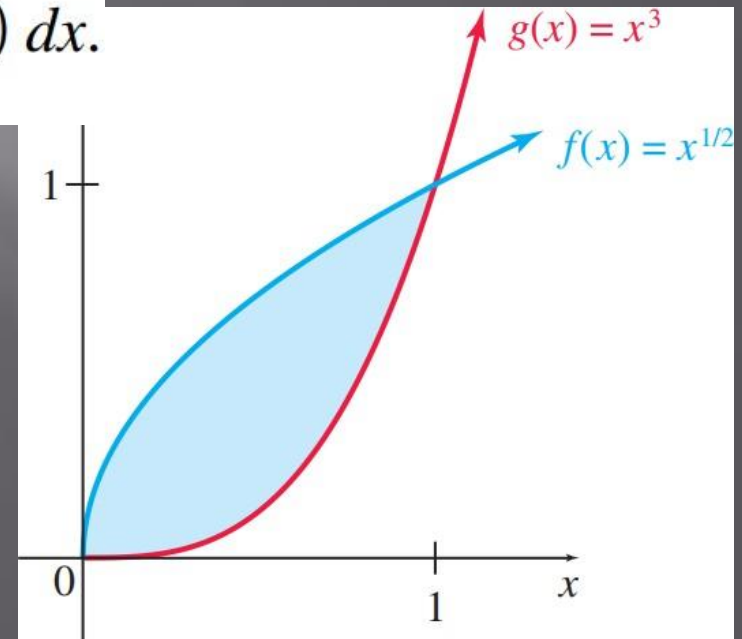


# Areas Between Curves (6/8)

## Example 2 (3/4)

Find the area between the curves  $f(x) = \sqrt{x}$  and  $g(x) = x^3$

$$\int_a^b [f(x) - g(x)] dx = \int_0^1 (x^{1/2} - x^3) dx.$$



# Areas Between Curves (6/8)

## Example 2 (4/4)

Find the area between the curves  $f(x) = \sqrt{x}$  and  $g(x) = x^3$

$$\begin{aligned}\int_a^b [f(x) - g(x)] dx &= \int_0^1 (x^{1/2} - x^3) dx \\&= \left( \frac{x^{3/2}}{3/2} - \frac{x^4}{4} \right) \Big|_0^1 \\&= \left( \frac{2}{3}x^{3/2} - \frac{x^4}{4} \right) \Big|_0^1 \\&= \frac{2}{3}(1) - \frac{1}{4} \\&= \frac{5}{12}.\end{aligned}$$

# Areas Between Curves (7/8)

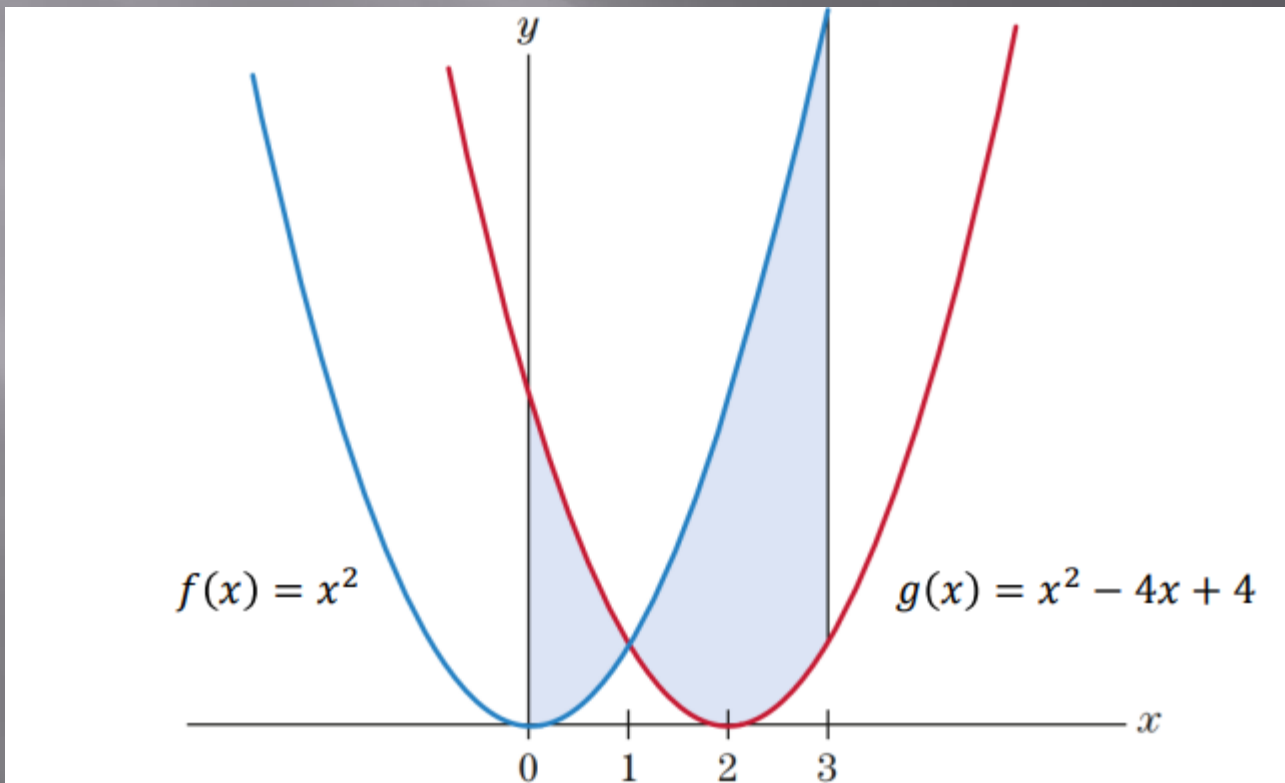
## Example 3 (1/4)

Find the area bounded by  $f(x) = x^2$ ,  $g(x) = (x - 2)^2$  from  $x = 0$  to  $x = 3$ .

# Areas Between Curves (7/8)

## Example 3 (2/4)

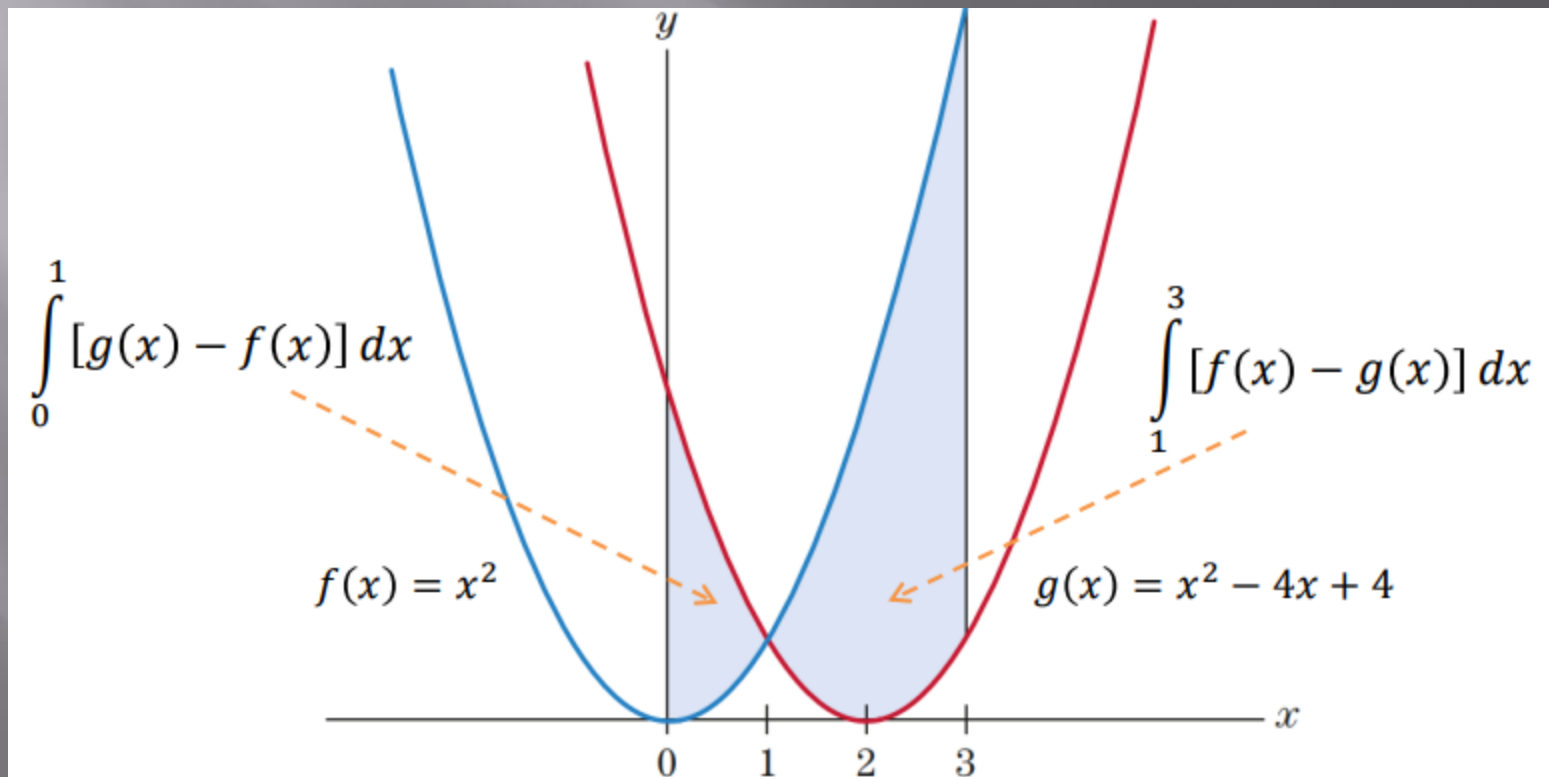
Find the area bounded by  $f(x) = x^2$ ,  $g(x) = (x - 2)^2$  from  $x = 0$  to  $x = 3$ .



# Areas Between Curves (7/8)

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# Areas Between Curves (7/8)

## Example 3 (3/4)

Find the area bounded by  $f(x) = x^2$ ,  $g(x) = (x - 2)^2$  from  $x = 0$  to  $x = 3$ .

$$\begin{aligned} [\text{Area from } x = 0 \text{ to } x = 1] &= \int_0^1 [(x^2 - 4x + 4) - (x^2)]dx \\ &= \int_0^1 (-4x + 4)dx \\ &= (-2x^2 + 4x) \Big|_0^1 = 2 - 0 = 2, \text{ and} \end{aligned}$$



# Areas Between Curves (7/8)

## Example 3 (4/4)

Find the area bounded by  $f(x) = x^2$ ,  $g(x) = (x - 2)^2$  from  $x = 0$  to  $x = 3$ .

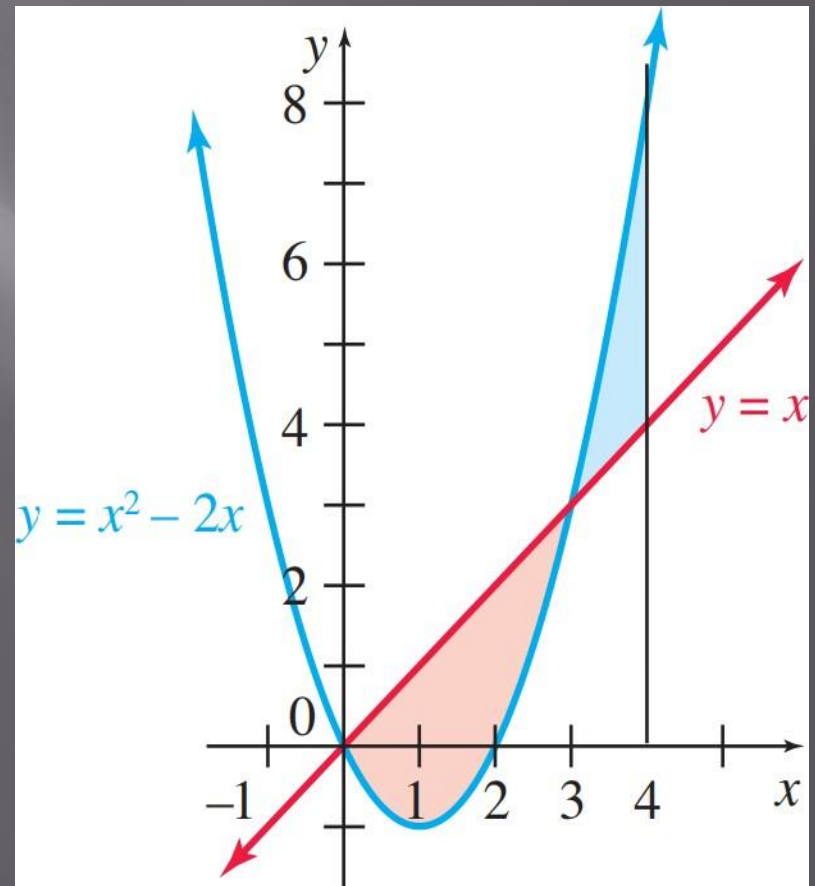
$$\begin{aligned} [\text{Area from } x = 1 \text{ to } x = 3] &= \int_1^3 [(x^2) - (x^2 - 4x + 4)]dx \\ &= \int_1^3 (4x - 4)dx \\ &= (2x^2 - 4x) \Big|_1^3 = 6 - (-2) = 8. \end{aligned}$$

Thus, the total area is  $2 + 8 = 10$ .

# Areas Between Curves (8/8)

## Example 4 (1/3)

Find the area of the region enclosed by  $y = x^2 - 2x$ ,  $y = x$  on  $[0, 4]$ .

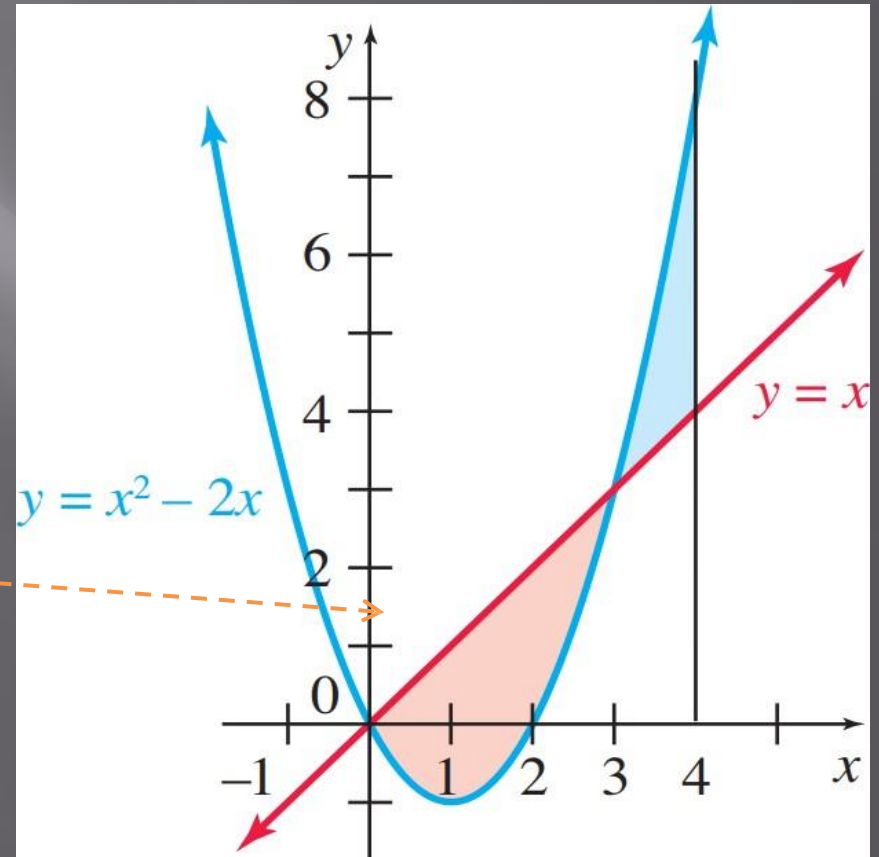


# Areas Between Curves (8/8)

## Example 4 (2/3)

Find the area of the region enclosed by  $y = x^2 - 2x$ ,  $y = x$  on  $[0, 4]$ .

$$\int_0^4 [x - (x^2 - 2x)] dx$$



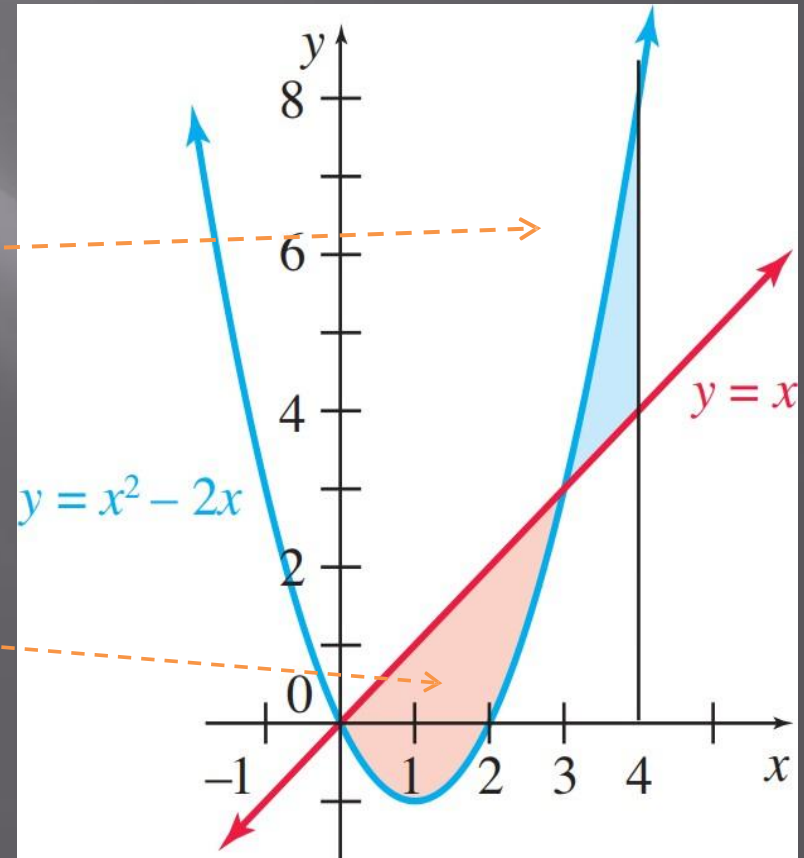
# Areas Between Curves (8/8)

## Example 4 (2/3)

Find the area of the region enclosed by  $y = x^2 - 2x$ ,  $y = x$  on  $[0, 4]$ .

$$\int_0^4 [(x^2 - 2x) - x] dx$$

$$\int_0^4 [x - (x^2 - 2x)] dx$$



# Areas Between Curves (8/8)

## Example 4 (3/3)

Find the area of the region enclosed by  $y = x^2 - 2x$ ,  $y = x$  on  $[0, 4]$ .

$$\begin{aligned}\text{Area} &= \int_0^3 [x - (x^2 - 2x)] dx + \int_3^4 [(x^2 - 2x) - x] dx \\&= \int_0^3 (-x^2 + 3x) dx + \int_3^4 (x^2 - 3x) dx \\&= \left( \frac{-x^3}{3} + \frac{3x^2}{2} \right) \Big|_0^3 + \left( \frac{x^3}{3} - \frac{3x^2}{2} \right) \Big|_3^4 \\&= \left( -9 + \frac{27}{2} - 0 \right) + \left( \frac{64}{3} - 24 - 9 + \frac{27}{2} \right) \\&= \frac{19}{3}\end{aligned}$$

# Thank You

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